

**Your grade: 96.66%**Your latest: **96.66%** • Your highest: **96.66%** • To pass you need at least 80%. We keep your highest score.

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1. This scenario is adapted from a real-world application, with specifics altered for confidentiality.

1 / 1 point



As a distinguished researcher in the City of Peacetopia, you face a unique challenge. The citizens are universally afraid of birds, and your task is to develop an algorithm to detect birds flying over the city and alert the populace.

The City Council has provided you with 10,000,000 sky images from Peacetopia's security cameras, labeled as follows:

- $y = 0$ : There is no bird on the image
- $y = 1$ : There is a bird on the image

**Your objective is to engineer an algorithm that can accurately classify new security camera images.** Decisions regarding the evaluation metric and data structuring into train/dev/test sets are critical.

The City Council has outlined their desires for the algorithm:

1. High accuracy.
2. Quick response time for classifying new images.
3. Low memory requirement to function on small processors across various cameras.

**True or False:** You realize that having a single evaluation metric will expedite development and ease algorithm comparison.

- ☐ False
- ☒ True

 **Nice work**

A singular focus on one evaluation metric can significantly speed up development and make it easier to compare and refine algorithms.

2. After further discussions, the city narrows down its criteria to:

1 / 1 point

- "We need an algorithm that can let us know a bird is flying over Peacetopia as accurately as possible."
- "We want the trained model to take no more than 10 seconds to classify a new image."
- "We want the model to fit in 10MB of memory."
- "We require a minimum of 98% test accuracy."

If you had the three following models, which one would you choose?

☐

| Test Accuracy | Runtime | Memory size |
|---------------|---------|-------------|
| 97%           | 1 sec   | 3MB         |

☐

| Test Accuracy | Runtime | Memory size |
|---------------|---------|-------------|
| 97%           | 3 sec   | 2MB         |

☒

| Test Accuracy | Runtime | Memory size |
|---------------|---------|-------------|
| 98%           | 9 sec   | 9MB         |

☐

| Test Accuracy | Runtime | Memory size |
|---------------|---------|-------------|
| 99%           | 13 sec  | 9MB         |

✔ Nice work

The runtime is less than 10 seconds, and the accuracy meets the minimum 98% requirement.

✔ Nice work

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3. Based on the context of a city's data analysis project, which of the following statements is true regarding the metrics used?

1 / 1 point

- ☐ Accuracy, running time, and memory size are all optimizing metrics because you want to do well on all three.
- ☐ Accuracy, running time, and memory size are all satisficing metrics because you have to do sufficiently well on all three for your system to be acceptable.
- ☒ Accuracy is an optimizing metric; running time and memory size are satisficing metrics.
- ☐ Accuracy is a satisficing metric; running time and memory size are an optimizing metric.

✔ Nice work

Accuracy is a metric we aim to maximize, while running time and memory size have acceptable thresholds.

4. Before implementing your algorithm, you need to split your data into train/dev/test sets. You have a dataset of 10,000,000 examples. Which of these do you think is the best choice?

1 / 1 point

|                       |           |           |           |
|-----------------------|-----------|-----------|-----------|
| <input type="radio"/> | Train     | Dev       | Test      |
|                       | 3,333,334 | 3,333,333 | 3,333,333 |

|                       |       |     |      |
|-----------------------|-------|-----|------|
| <input type="radio"/> | Train | Dev | Test |
|                       |       |     |      |

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1 / 1 point

☐

| Train     | Dev       | Test      |
|-----------|-----------|-----------|
| 3,333,334 | 3,333,333 | 3,333,333 |

☐

| Train     | Dev       | Test      |
|-----------|-----------|-----------|
| 6,000,000 | 3,000,000 | 1,000,000 |

☒

| Train     | Dev     | Test    |
|-----------|---------|---------|
| 9,500,000 | 250,000 | 250,000 |

☐

| Train     | Dev       | Test      |
|-----------|-----------|-----------|
| 6,000,000 | 1,000,000 | 3,000,000 |

✔ Nice work

With a large dataset, smaller dev and test sets are sufficient for evaluating bias and variance.

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5. Now that you've set up your train/dev/test sets, the City Council comes across another 1,000,000 images from social media and offers them to you. These images have a different distribution from the images the City Council originally provided, but you think they could help your algorithm. Should you add this data to the training set?

1 / 1 point

☒ Yes.☐ No. **Nice work**

This will cause the training and dev/test set distributions to become different. However, as long as the dev/test distributions are the same, you are aiming at the same target.

6. One member of the City Council wants to add 1,000,000 citizen data images evenly to the training, development (dev), and test sets. Your original data is from security cameras, and you object because:

1 / 1 point

☒ If we add the images to the test set, then it won't reflect the distribution of data (security cameras) expected in production.☐ The additional data would significantly slow down training time.☐ The training set will not be as accurate because of the different distributions.☐ The 1,000,000 citizen data images do not have a consistent input-output relationship as the security camera data. **Nice work**

The test set must accurately represent the real-world data distribution to properly evaluate model performance. Adding citizen data images to the test set would skew this distribution.

7. Human performance for identifying birds is < 1%, training set error is 5.2%, and dev set error is 7.3%.

1 / 1 point

Which of the options below is the best next step?

- ☐ Try an ensemble model to reduce bias and variance.
- ☐ Validate the human data set with a sample of your data to ensure the images are of sufficient quality.
- ☐ Get more data or apply regularization to reduce variance.
- ☒ Train a bigger network to reduce the 5.2% training error.

✔ Nice work

Avoidable bias is 4.2%, which is larger than the 2.1% variance, so reducing bias is the priority.

8. You ask a few people to label a bird species dataset to determine human-level performance. The following error rates were recorded:

1 / 1 point

|                                               |            |
|-----------------------------------------------|------------|
| Bird watching expert #1                       | 0.3% error |
| Bird watching expert #2                       | 0.5% error |
| Normal person #1 (not a bird watching expert) | 1.0% error |
| Normal person #2 (not a bird watching expert) | 1.2% error |

If your goal is to use "human-level performance" as an estimate for Bayes error, how would you define "human-level performance" in this scenario?

- ☒ 0.3% (The lowest error rate achieved by an expert)
- ☐ 0.75% (Average of all four error rates)
- ☐ 0.4% (Average of the two experts' error rates)
- ☐ 0.0% (Perfect accuracy, representing an unattainable ideal)

✔ Nice work

The best performance of an expert is the closest practical estimate of Bayes error.



9. Which of the following statements do you agree with?

1 / 1 point

- ☐ A learning algorithm's performance can never be better than human-level performance nor better than Bayes error.
- ☒ A learning algorithm's performance can be better than human-level performance but it can never be better than Bayes error.
- ☐ A learning algorithm's performance can be better than human-level performance and better than Bayes error.
- ☐ A learning algorithm's performance can never be better than human-level performance but it can be better than Bayes error.

✔ Nice work

This statement accurately reflects the relationship between the learning algorithm, human-level performance, and Bayes error.

10. Which of the following best describes the **most effective next step in your project**, given the following performance metrics?

1 / 1 point

- Human-level performance: 0.1%
  - Training set error: 2.0%
  - Dev set error: 2.1%
- ☒ Prioritize actions to decrease bias by increasing model complexity, as the training error significantly exceeds human-level performance.
  - ☐ Deploy the model to target devices to evaluate against satisficing metrics.
  - ☐ Continue tuning until the training set error matches human-level performance, focusing solely on the optimizing metric.
  - ☐ Evaluate the test set to determine the variance.

✔ Nice work

Yes, addressing the largest performance gap (between human-level and training error) is the most efficient strategy.



11. After running your model with the test set, you find the error rate is 7.0% compared to a 2.1% error rate for the dev set and 2.0% for the training set. What can you conclude? (Choose all that apply)

1 / 1 point

☐ You have underfitted to the dev set.

☒ You have overfitted to the dev set.

✔ Nice work

The significantly higher test set error compared to the dev set indicates overfitting to the dev set.

☒ You should try to get a bigger dev set.

✔ Nice work

A larger dev set can provide a more reliable performance estimate and help address overfitting issues.

☐ Try decreasing regularization for better generalization with the dev set.

12. After working on this project for a year, you finally achieve: Human-level performance, 0.10%, Training set error, 0.05%, Dev set error, 0.05%. Which of the following are true? (Check all that apply.)

0.8 / 1 point

☐ With only 0.05% further progress to make, you should quickly be able to close the remaining gap to 0%.

☒ All or almost all of the avoidable bias has been accounted for.

✔ Nice work

Exceeding human performance strongly suggests that most, if not all, significant sources of avoidable bias have been addressed.

☐ It is highly unlikely this result is purely a statistical anomaly, but statistical noise may still contribute to the error.

☐ You are close to Bayes error and there is a possibility of overfitting.

⊗ You didn't select all the correct answers

13. Your system is now very accurate but has a higher false negative rate than the City Council of Peacetopia

1 / 1 point



⊗ You didn't select all the correct answers

13. Your system is now very accurate but has a higher false negative rate than the City Council of Peacetopia would like. What is **your best** next step?

1 / 1 point

- ☒ Reset your “target” (metric) for the team and tune to it.
- ☐ Look at all the models you’ve developed during the development process and find the one with the lowest false negative error rate.
- ☐ Expand your model size to account for more corner cases.
- ☐ Pick false negative rate as the new metric, and use this new metric to drive all further development.

✓ Nice work

The target has shifted so an updated metric is required.

14. Over the last few months, a new species of bird has been slowly migrating into the area, so the performance of your system slowly degrades because your data is being tested on a new type of data. There are only 1,000 images of the new species. The city expects a better system from you within the next 3 months.

1 / 1 point

**Which of these should you do first?**

- ☒ Augment your data to increase the number of images of the new bird species.
- ☐ Split them between dev and test and re-tune.
- ☐ Put the new species' images in training data to learn their features.
- ☐ Add pooling layers to downsample features to accommodate the new species.

✓ Nice work

Generating more training examples through data augmentation is the most effective first step to address the limited data for the new species.

15. The City Council thinks that having more cats in the city would help scare off birds. They are so happy with your work on the Bird detector that they also hire you to build a Cat detector.

0.8 / 1 point

Because of years of working on Cat detectors, you have such a huge dataset of 100,000,000 cat images that training on this data takes about two weeks.

✔ Nice work

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Which of the statements do you agree with? (Check all that agree.)

- ☒ Buying faster computers could speed up your team's iteration speed and thus your team's productivity.

✔ Nice work

Enhanced computational resources can reduce training time and improve productivity.

- ☐ Having built a good Bird detector, you should be able to take the same model and hyperparameters and just apply it to the Cat dataset, so there is no need to iterate.
- ☐ If 100,000,000 examples is enough to build a good enough Cat detector, you might be better off training with just 10,000,000 examples to gain a ~10x improvement in how quickly you can run experiments, even if each model performs a bit worse because it's trained on less data.
- ☒ Needing two weeks to train will limit the speed at which you can iterate.

✔ Nice work

The long training time constrains how quickly you can test and refine models.

⊗ You didn't select all the correct answers